

Cambridge (CIE) IGCSE Additional Maths



Your notes

Logarithmic & Exponential Functions

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Your notes

Exponential Functions

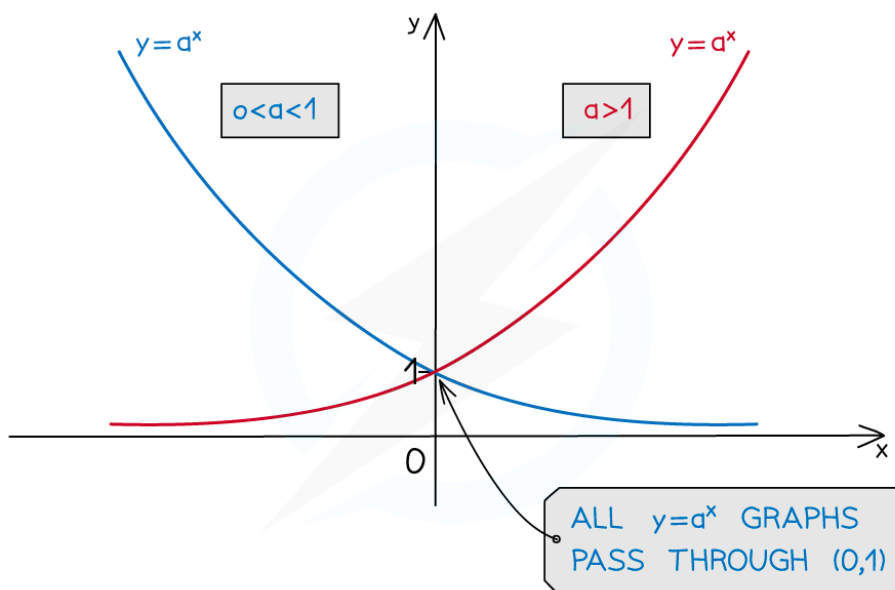
Exponential Functions

What is an exponential function?

- An exponential function is a function where the variable is the power
- They are of the form $y = a^x$ with $a > 0$

What is an exponential graph?

- All graphs of the form $y = a^x$ will pass through $(0, 1)$ because $a^0 = 1$
- The x -axis is an **asymptote**



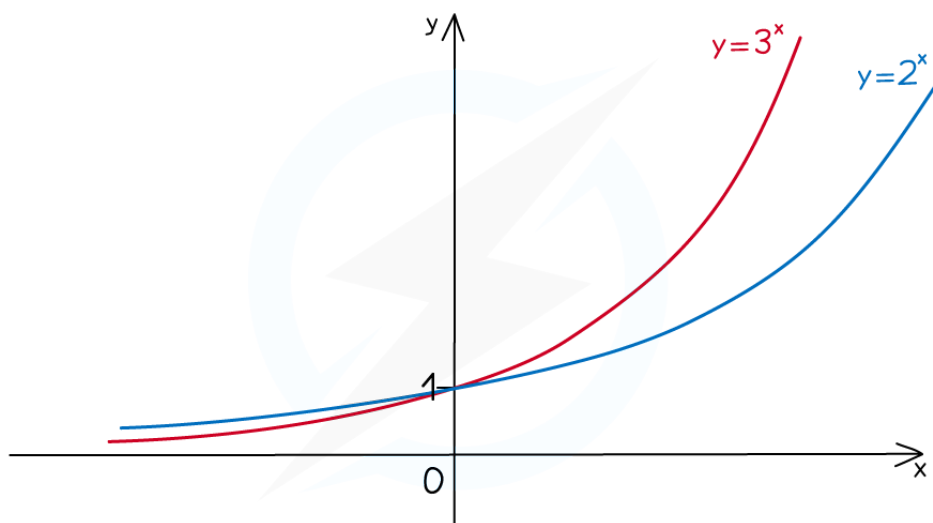
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Exponential graphs when $a > 1$

- Where $x < 0$ the higher value of a is the "lower" graph
- Where $x > 0$ the higher value of a is the "higher" graph
- $a > 1$ is **exponential growth**



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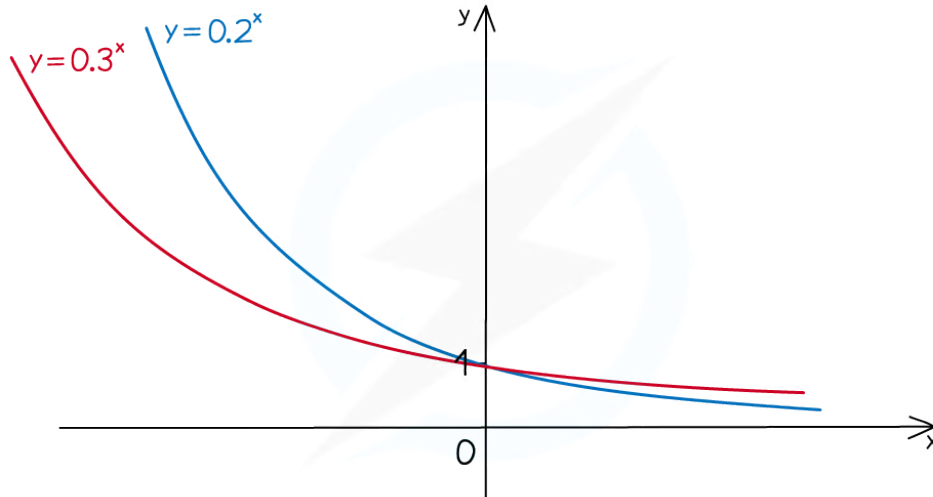
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What about when $a \leq 1$?

- You may like to think about why $a = 1$ is **not** considered... If $a = 1$, $y = 1^x = 1$ for all values of x
- $0 < a < 1$ represents **exponential decay**
 - Where $x < 0$ the higher value of a is the “higher” graph
 - Where $x > 0$ the higher value of a is the “lower” graph



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Worked Example



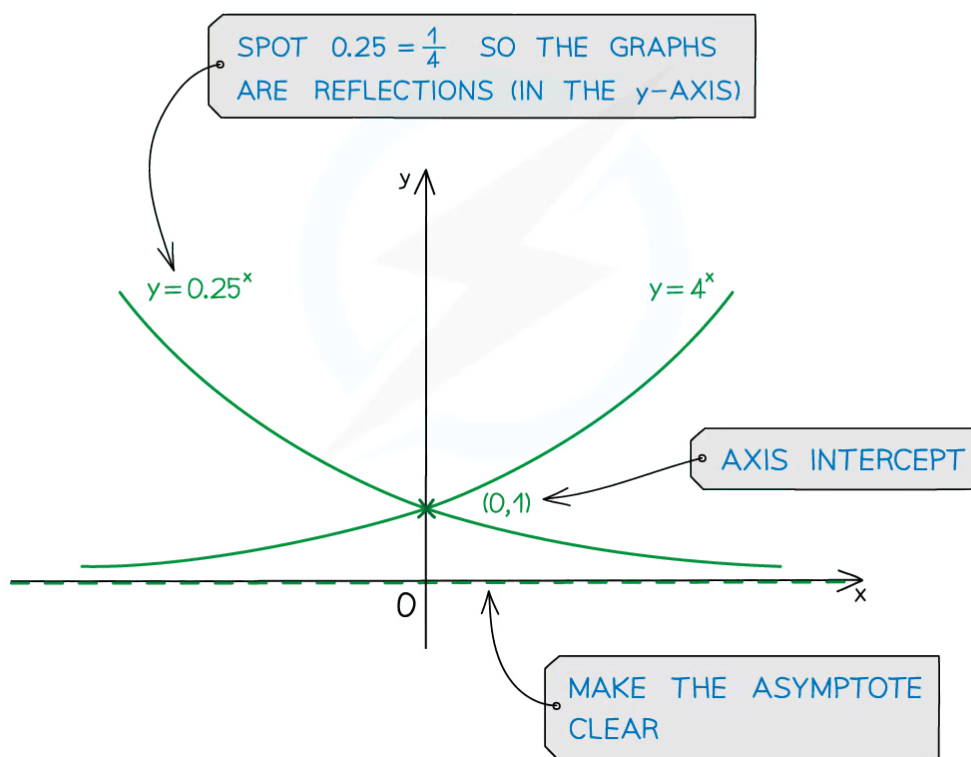
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On the same diagram sketch the graphs of

$$y = 4^x \text{ and } y = 0.25^x.$$

Mark any points where the graphs cross the coordinate axes and make any asymptotes clear.



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"e"

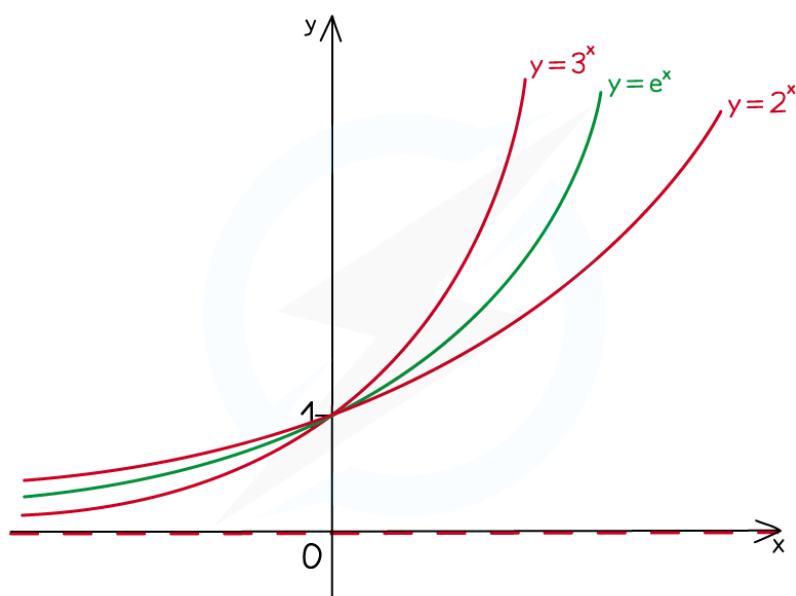
What is e, the exponential function?

- The exponential function is $y = e^x$

- e is an irrational number
- $e \approx 2.718$
- As with other exponential graphs $y = e^x$
 - passes through $(0, 1)$
 - has the x -axis as an asymptote



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What is the big deal with e ?

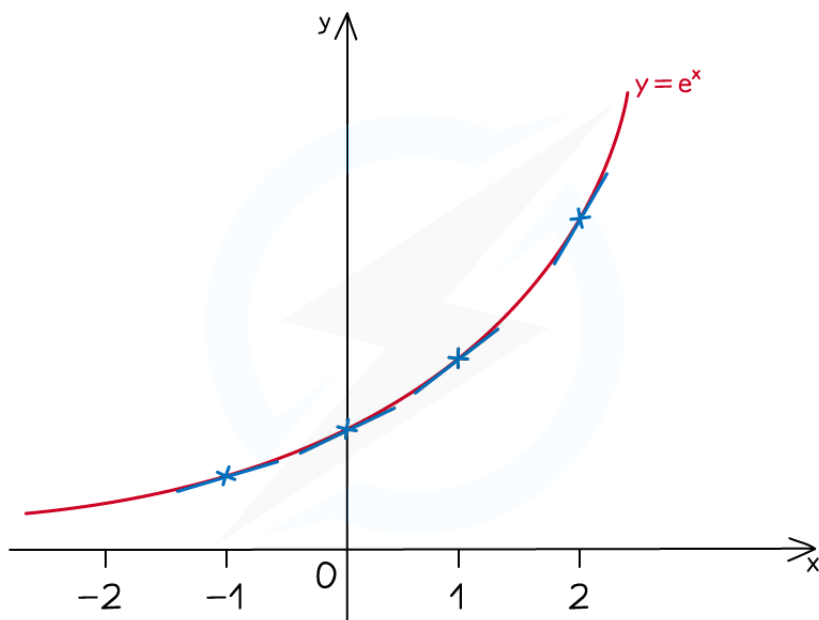
- $y = e^x$ has the particular property

$$\frac{dy}{dx} = e^x$$

- i.e. for every real number x , the gradient of $y = e^x$ is also equal to e^x (see Differentiating e^x and $\ln x$)



Your notes



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x	y	dy / dx
-1	0.3678...	0.3678...
0	1	1
1	2.7182...	2.7182...
2	7.3890...	7.3890...

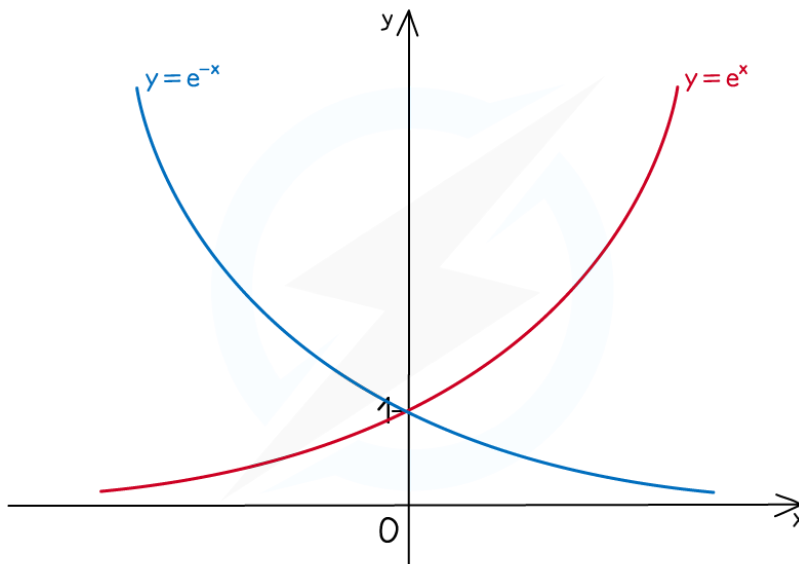
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The negative exponential graph

- $y = e^{-x}$ is a reflection in the y -axis of $y = e^x$



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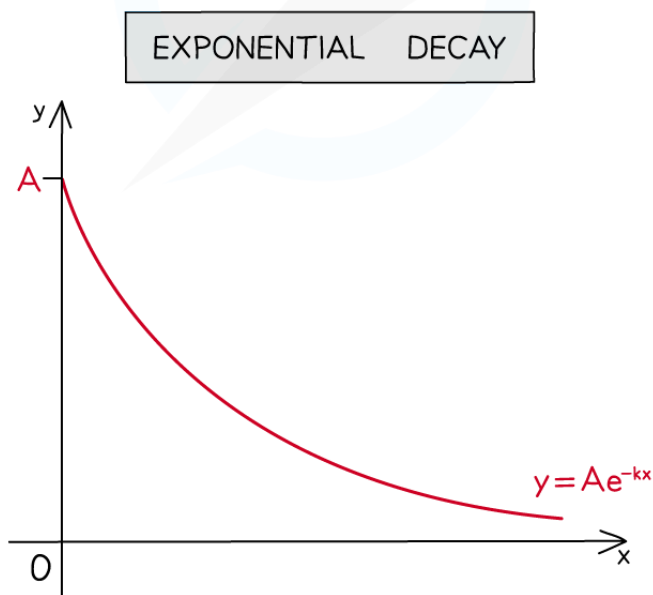
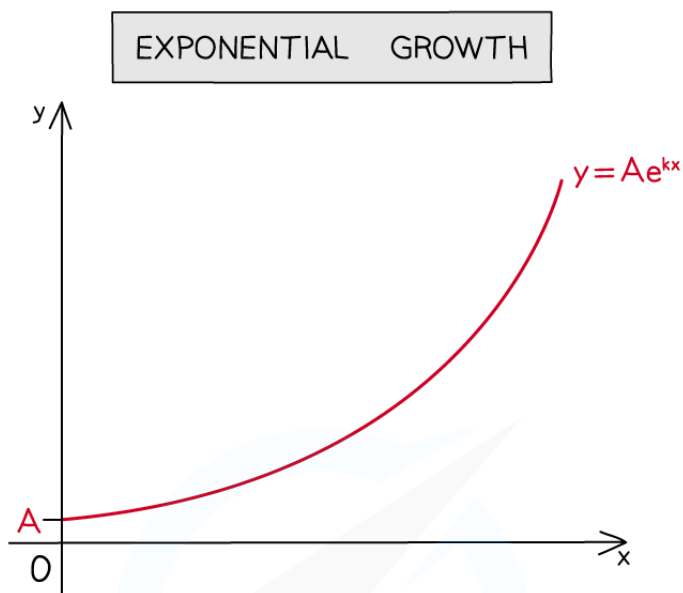


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What is exponential growth and decay?



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- $y = Ae^{kx}$ ($k > 0$) is exponential growth
- $y = Ae^{-kx}$ ($k > 0$) is exponential decay

- **A** is the initial value
- **k** is a (usually positive) constant
- A **negative sign** is used in the equation making clear whether it is growth or decay



Worked Example



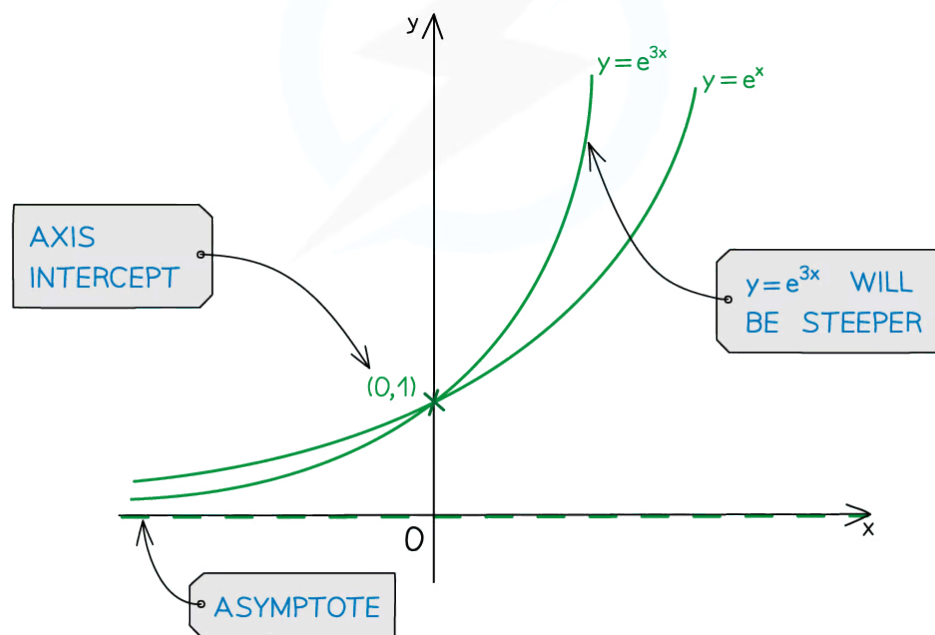
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- ?** (a) On the same diagram sketch the graphs of
- (i) $y = e^x$
 - (ii) $y = e^{3x}$
- (b) On the same diagram sketch the graphs of
- (i) $y = e^{-x}$
 - (ii) $y = e^{-7x}$

a) (i) & (ii)

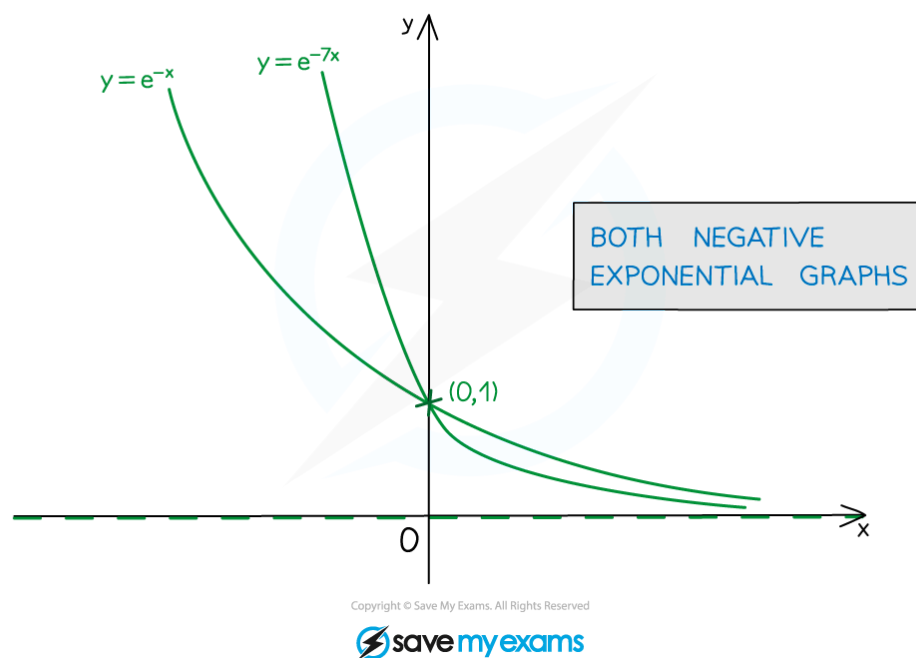


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b) (i) & (ii)





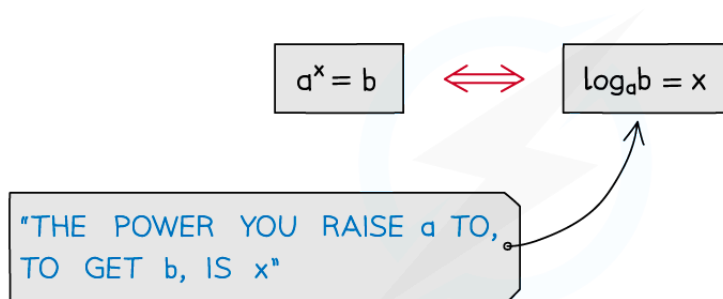
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Logarithmic Functions

Logarithmic Functions

What are logarithmic functions?

- A logarithm is the inverse of raising to a power
- If $a = b^x$ then $\log_b a = x$
 - $a > 0$
 - b is called the **base** of the logarithm
- Try to get used to 'reading' logarithm statements to yourself
 - $\log_{\dots}(\dots) =$ would be read as "the power that you raise ... to, to get ..., is"
 - So $\log_5(125) = 3$ would be read as "the power that you raise 5 to, to get 125, is 3"



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- A logarithm is the inverse of raising to a power so we can use rules to simplify logarithmic functions



Your notes

$$\log_a(a^x) = x$$

"THE POWER YOU RAISE
a TO, TO GET a^x , IS x"

$$a^{\log_a x} = x$$

"log" AND "RAISING
TO A POWER" ARE
INVERSES

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Why use logarithms?

- Logarithms allow us to solve equations where the exponent is the unknown value
 - We can solve some of these by inspection
 - For example, for the equation $2^x = 8$ we know that x must be 3
 - Logarithms allow use to solve more complicated problems
 - For example, the equation $2^x = 10$ does not have a clear answer
 - Instead, we can use our calculator to find the value of $\log_2 10$

How do I use logarithms?



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e.g. WRITE $\frac{2^{\log_3 81} + 3^{\log_{10} 1000}}{5^{\log_5 5}}$ IN THE FORM $\frac{a}{b}$
WHERE a AND b ARE INTEGERS TO BE FOUND.

$2^{\log_3 81} = 2^4 = 16$
"THE POWER YOU RAISE 3 TO, TO GET 81, IS 4"

$3^{\log_{10} 1000} = 3^3 = 27$
"THE POWER YOU RAISE 10 TO, TO GET 1000, IS 3"

$5^{\log_5 5} = 5^1 = 5$
 $\log_5 5 = \log_5 5^1 = 1$

$$\therefore \frac{2^{\log_3 81} + 3^{\log_{10} 1000}}{5^{\log_5 5}} = \frac{16 + 27}{5}$$

$$= \frac{43}{5}$$

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- Recognising the rules of logarithms allows expressions to be simplified



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e.g. WITHOUT USING A CALCULATOR,
EVALUATE $\log_2 64\sqrt{2}$

RECOGNISE 64 AND $\sqrt{2}$ ARE POWERS OF 2

LET $x = \log_2 64\sqrt{2}$

"THE POWER YOU RAISE
2 TO, TO GET $64\sqrt{2}$, IS x"

$2^x = 64\sqrt{2}$

REWRITE THE STATEMENT

$2^x = 2^6 \times 2^{\frac{1}{2}}$

$2^x = 2^{6+\frac{1}{2}}$

USING LAWS
OF INDICES

$2^x = 2^{\frac{13}{2}}$

$x = \frac{13}{2}$

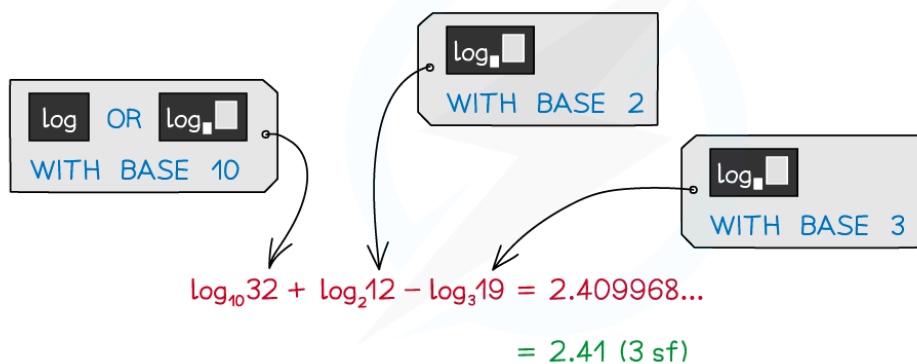
$\therefore \log_2 64\sqrt{2} = \frac{13}{2}$

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- Recognition of common powers helps in simple cases
 - Powers of 2: $2^0 = 1, 2^1 = 2, 2^2 = 4, 2^3 = 8, 2^4 = 16, \dots$
 - Powers of 3: $3^0 = 1, 3^1 = 3, 3^2 = 9, 3^3 = 27, 3^4 = 81, \dots$
 - The first few powers of 4, 5 and 10 should also be familiar

For more awkward cases a calculator is needed

e.g. USE A CALCULATOR TO EVALUATE $\log_{10}32 + \log_212 - \log_319$
GIVING YOUR ANSWER TO 3 SIGNIFICANT FIGURES



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- Calculators can have, possibly, three different logarithm buttons



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- This button allows you to type in any number for the base



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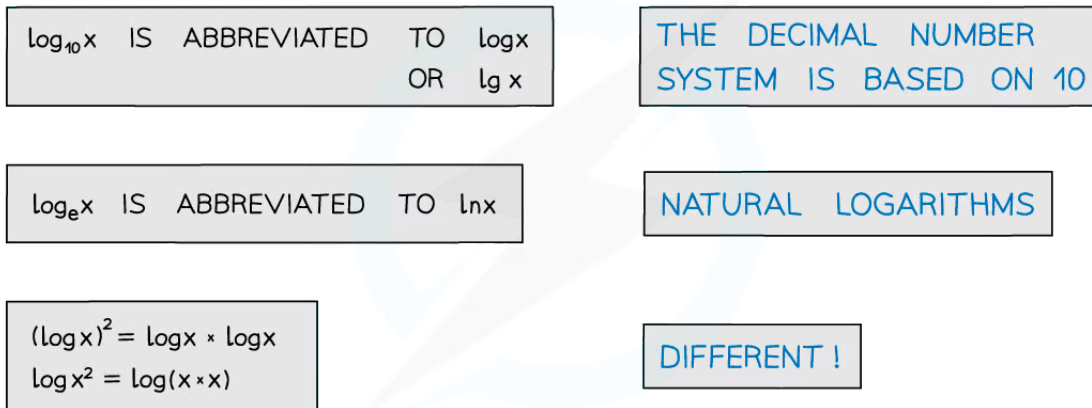
- Natural logarithms (see “e”)



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- Shortcut for base 10 although SHIFT button needed
- Before calculators, logarithmic values had to be looked up in printed tables

What notation might I see with logarithms?



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- 10 is a common base
 - $\log_{10} x$ is abbreviated to $\log x$ or $\lg x$
- The value **e** is another common base
 - $\log_e x$ is abbreviated to $\ln x$
- $(\log x)^2 \neq \log x^2$



Examiner Tips and Tricks



Your notes

- Before going into the exam, make sure you are completely familiar with your calculator and know how to use its logarithm functions

$\ln x$

What is $\ln$?

- $\ln$ is a function that stands for **natural logarithm**
- It is a logarithm where the **base** is the constant "**e**"
 - $\ln x \equiv \log_e x$
 - It is important to remember that $\ln$ is a function and **not a number**
- The natural logarithm ($\ln x$) and the exponential function (e^x) are inverses of each other
- It is defined for all positive numbers ($x > 0$)
 - $\ln x$ cannot be defined for negative numbers or $x = 0$

What are the properties of $\ln$?

- Using the **definition** of a **logarithm** you can see
 - $\ln 1 = 0$
 - $\ln e = 1$
 - $\ln e^x = x$
 - $\ln x$ is only defined for positive x

How can I solve equations involving e & $\ln$?

- The functions e^x and $\ln x$ are **inverses** of each other
 - If $e^x = a$ then $x = \ln a$
 - If $\ln x = a$ then $x = e^a$
 - If $e^{f(x)} = g(x)$ then $f(x) = \ln g(x)$
 - If $\ln f(x) = g(x)$ then $f(x) = e^{g(x)}$
- If your equation involves " e " then try to get all the " e " terms on one side



Your notes

- If "e" terms are multiplied, you can add the powers
 - $e^x \times e^y = e^{x+y}$
 - You can then apply $\ln$ to both sides of the equation
- If "e" terms are added, try transforming the equation with a substitution
 - For example: If $y = e^x$ then $e^{4x} = y^4$
 - You can then solve the resulting equation (usually a quadratic)
 - Once you solve for y then solve for x using the substitution formula
- If your equation involves " $\ln$ ", try to combine all " $\ln$ " terms together
 - Use the laws of logarithms to combine terms into a single term
 - If you have $\ln f(x) = \ln g(x)$ then solve $f(x) = g(x)$
 - If you have $\ln f(x) = k$ then solve $f(x) = e^k$



Examiner Tips and Tricks

- If you're working on the non-calculator exam paper you may need to leave answers as exact values so using $\ln x$ is a good idea
 - This can be solved with e^x which can then be left as an exact value



Worked Example

Solve the equation $e^{2x} = 5$, leaving your answer as an exact value.

Take the natural logarithm of both sides.

$$\ln(e^{2x}) = \ln 5$$

Use the property that $\ln(e^a) = a$.

$$2x = \ln 5$$

Divide both sides by 2.

$$x = \frac{\ln 5}{2}$$

Do not use your calculator to evaluate this as the question asks for the answer given as an exact value.

$$\mathbf{x = \frac{\ln 5}{2}}$$



Your notes



Your notes

Laws of Logarithms

Laws of Logarithms

What are the laws of logarithms?

- Laws of logarithms allow you to simplify and manipulate expressions involving logarithms
 - The laws of logarithms are equivalent to the **laws of indices**
- The laws you need to know are, given $a, x, y > 0$:
 - $\log_a xy = \log_a x + \log_a y$
 - This relates to $a^x \times a^y = a^{x+y}$
 - $\log_a \frac{x}{y} = \log_a x - \log_a y$
 - This relates to $a^x \div a^y = a^{x-y}$
 - $\log_a x^m = m \log_a x$
 - This relates to $(a^x)^y = a^{xy}$

$$\log_a xy = \log_a x + \log_a y$$

$$\text{RELATES TO } a^x \times a^y = a^{x+y}$$

$$\log_a \left(\frac{x}{y} \right) = \log_a x - \log_a y$$

$$\text{RELATES TO } \frac{a^x}{a^y} = a^{x-y}$$

$$\log_a x^k = k \log_a x$$

$$\text{RELATES TO } (a^x)^y = a^{xy}$$

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- There are also some particular results these lead to



Your notes

- $\log_a a = 1$
- $\log_a a^x = x$
- $a^{\log_a x} = x$
- $\log_a 1 = 0$
- $\log_a \left(\frac{1}{x} \right) = -\log_a x$

$$\log_a a = 1$$

"THE POWER YOU RAISE
a TO, TO GET a, IS 1"

$$\log_a a^x = x$$

$$\log_a a^x = x \log_a a$$

$$= x$$

$$a^{\log_a x} = x$$

AN OPERATION AND
ITS INVERSE

$$\log_a 1 = 0$$

$$a^0 = 1$$

$$\log_a \frac{1}{x} = -\log_a x$$

$$\log_a \frac{1}{x} = \log_a x^{-1}$$

$$= -\log_a x$$

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- Beware...
 - ... $\log_a (x + y) \neq \log_a x + \log_a y$
- These results apply to $\ln x$ ($\log_e x$) too
 - Two particularly useful results are



Your notes

- $\ln e^x = x$

- $e^{\ln x} = x$

How do I use the laws of logarithms?

- Laws of logarithms can be used to ...
 - ... simplify expressions
 - ... solve logarithmic equations
 - ... solve exponential equations

e.g. WRITE $3\log_2(2x+3) + \log_2 5 - 2\log_2(x+1)$
AS A SINGLE LOGARITHM

$$3\log_2(2x+3) + \log_2 5 - 2\log_2(x+1) = \log_2(2x+3)^3 + \log_2 5 - \log_2(x+1)^2$$

$$k\log_a x = \log_a x^k$$

$$= \log_2 5(2x+3)^3 - \log_2(x+1)^2$$

$$\log_a x + \log_a y = \log_a(xy)$$

$$= \log_2 \frac{5(2x+3)^3}{(x+1)^2}$$

$$\log_a x - \log_a y = \log_a \frac{x}{y}$$

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Examiner Tips and Tricks

- Remember to check whether your solutions are valid
 - $\log(x+k)$ is only defined if $x > -k$
 - You will lose marks if you forget to reject invalid solutions

**Worked Example**

a) Write the expression $2 \log 4 - \log 2$ in the form $\log k$, where $k \in \mathbb{Z}$.

Using the law $\log_a x^m = m \log_a x$ we can rewrite $2 \log 4$ as $\log 4^2 = \log 16$.

$$2 \log 4 - \log 2 = \log 16 - \log 2$$

Using the law $\log_a \frac{x}{y} = \log_a x - \log_a y$:

$$\log 16 - \log 2 = \log \frac{16}{2} = \log 8$$

$$2 \log 4 - \log 2 = \log 8$$

b) Hence, or otherwise, solve $2 \log 4 - \log 2 = -\log \frac{1}{x}$.

Rewrite the equation using the expression found in part (a).

$$2 \log 4 - \log 2 = \log 8:$$

$$\log 8 = -\log \frac{1}{x}$$

Using the index law $\frac{1}{x} = x^{-1}$:

$$\log 8 = -\log x^{-1}$$



Your notes



Your notes

Using the law $\log_a x^m = m \log_a x$:

$$\log 8 = \log x^{-(-1)} = \log x$$

Compare the two sides.

$$\log 8 = \log x$$

$$x = 8$$

Change of Base

How do I change the base of a logarithm?

- The formula for changing the base of a logarithm is

$$\log_a x = \frac{\log_b x}{\log_b a}$$

- The value you choose for b does not matter, however if you do not have a calculator, you can choose b such that the problem will be possible to solve

Why change the base of a logarithm?

- The laws of logarithms can only be used if the logs have the same **base**
 - If a problem involves logarithms with different bases, you can change the base of the logarithm and then apply the laws of logarithms
- **Changing the base** of a logarithm can be particularly useful if you need to evaluate a log problem **without a calculator**
 - Choose the base such that you would know how to solve the problem from the equivalent exponent
- This formula had more use when calculators were less advanced
 - Some old calculators only had a button for logarithm of base 10
 - To calculate $\log_5 7$ on these calculators you would have to enter



Your notes

$$\frac{\log_{10} 7}{\log_{10} 5}$$

- The formula can be useful when evaluating a logarithm where the two numbers are **powers of a common number**

$$\log_4 8 = \frac{\log_2 8}{\log_2 4} = \frac{3}{2}$$

- The formula can be useful when you are solving equations and two logarithms have **different bases**

- For example, if you have $\log_3 k$ and $\log_9 n$ within the same equation

- You can rewrite $\log_9 n$ as $\frac{\log_3 n}{\log_3 9}$ which simplifies to $\frac{1}{2} \log_3 n$

- Or you can rewrite $\log_3 k$ as $\frac{\log_9 k}{\log_9 3}$ which simplifies to $2 \log_9 k$

- The formula also allows you to derive and use a formula for switching the numbers:

$$\log_a x = \frac{1}{\log_x a}$$

- Using the fact that $\log_x x = 1$



Examiner Tips and Tricks

- It is very rare that you will need to use the change of base formula
- Only use it when the bases of the logarithms are different



Worked Example



Your notes

By choosing a suitable value for b , use the change of base law to find the value of $\log_8 32$ without using a calculator.

Note that 8 and 32 are both powers of 2, where $8 = 2^3$ and $32 = 2^5$.

Therefore we can choose $b = 2$ in the change of base formula.

$$\log_a x = \frac{\log_b x}{\log_b a}$$

$$\log_8 32 = \frac{\log_2 32}{\log_2 8}$$

If $2^3 = 8$ then $\log_2 8 = 3$ and if $2^5 = 32$ then $\log_2 32 = 5$.

$$\frac{\log_2 32}{\log_2 8} = \frac{5}{3}$$

$$\log_8 32 = \frac{5}{3} = 1\frac{2}{3}$$



Your notes

Exponential Equations

Solving Exponential Equations

What are exponential equations?

- An exponential equation is an equation where the unknown is a power
 - In simple cases the solution can be spotted without the use of a calculator
 - For example,

$$5^{2x} = 125$$

$$2x = 3$$

$$x = \frac{3}{2}$$

- In more complicated cases the laws of logarithms should be used to solve exponential equations
- The **change of base** law can be used to solve some exponential equations without a calculator
 - For example,

$$27^x = 9$$

$$x = \log_{27} 9$$

$$= \frac{\log_3 9}{\log_3 27}$$

$$= \frac{2}{3}$$

How do we use logarithms to solve exponential equations?

- An exponential equation can be solved by taking logarithms of both sides
- The **laws of indices** may be needed to rewrite the equation first
- The **laws of logarithms** can then be used to solve the equation
 - **ln (log_e)** is often used
 - The answer is often written in terms of ln



Your notes

- A question may ask you to give your answer in a particular form
- Follow these steps to solve exponential equations
 - **STEP 1:** Take logarithms of both sides
 - **STEP 2:** Use the laws of logarithms to remove the powers
 - **STEP 3:** Rearrange to isolate x
 - **STEP 4:** Use logarithms to solve for x

What about hidden quadratics?

- Look for hidden squared terms that could be changed to form a quadratic
 - In particular look out for terms such as
 - $4^x = (2^2)^x = 2^{2x} = (2^x)^2$
 - $e^{2x} = (e^2)^x = (e^x)^2$



Examiner Tips and Tricks

- Always check which form the question asks you to give your answer in, this can help you decide how to solve it
- If the question requires an exact value you may need to leave your answer as a logarithm



Worked Example

Solve the equation $4^x - 3(2^{x+1}) + 9 = 0$. Give your answer correct to three significant figures.

'Spot' the hidden quadratic by noticing that $4^x = (2^2)^x = (2^x)^2$.

Rewrite the first term as a power of 2.

$$(2^x)^2 - 3(2^{x+1}) + 9 = 0$$

Rewrite the middle terms using the laws of indices: If $2^{x+1} = 2^x \times 2^1 = 2(2^x)$

$$(2^x)^2 - 3 \times 2(2^x) + 9 = 0$$

$$(2^x)^2 - 6(2^x) + 9 = 0$$



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Using a substitution can make this easier to solve.

$$\text{Let } u = 2^x$$

$$(u)^2 - 6(u) + 9 = 0$$

Factorise.

$$u^2 - 6u + 9 = 0$$

$$(u - 3)(u - 3) = 0$$

Solve to find u and substitute 2^x back in.

$$u = 3$$

$$2^x = 3$$

Solve the exponential equation $2^x = 3$ by taking logarithms of both sides.

$$\ln 2^x = \ln 3$$

Bring the power down using the law of logs $\ln x^m = m \ln x$.

$$x \ln 2 = \ln 3$$

Rearrange and solve.

$$x = \frac{\ln 3}{\ln 2} = 1.584...$$

$$\mathbf{x = 1.58 \text{ (3 s.f.)}}$$



Your notes

Transforming Relationships to Linear Form

Transforming Relationships in the Form $y = ax^n$

How do I use logarithms to linearise a graph in the form $y = ax^n$?

- Logarithms can be used to linearise graphs of **power functions**
- Suppose $y = ax^n$
 - You can take logarithms of both sides
 - $\ln y = \ln(ax^n)$
 - You can split the right hand side into the sum of two logarithms
 - $\ln y = \ln a + \ln(x^n)$
 - You can bring down the power in the final term
 - $\ln y = \ln a + n \ln x$
- $\ln y = \ln a + n \ln x$ is in linear form $Y = mX + c$
 - $Y = \ln y$
 - $X = \ln x$
 - $m = n$
 - $c = \ln a$

How can I use linearised form to find the unknown constants?

- After linearising the function it will be in the form $\ln y = \ln a + n \ln x$
 - n is the gradient of the straight line graph
 - $\ln a$ is the y -intercept of the straight line graph
- Once you know the value of the gradient of the straight line graph this is the value of n
- You will need to find the value of a by solving the equation $\ln a = c$



Your notes



Examiner Tips and Tricks

- You may need to leave your answer in exact form, especially in the non-calculator paper
 - e.g. to solve $\ln a = 2$ the answer will be $a = e^2$



Worked Example

The heights, h metres, and the amount of time spent sleeping, t hours, of a group of young giraffes can be modelled using $t = ah^b$, where a and b are constants.

The graph of $\ln h$ against $\ln t$ is a straight line passing through the points (1, -0.9) and (4, -4.5).

Find the values of a and b , giving your answers in exact form.

Find the gradient of the straight line between the two coordinates.

$$m = \frac{-4.5 - -0.9}{4 - 1} = -\frac{3.6}{3} = -1.2$$

Substitute $m = -1.2$, $x = \ln h$ and $y = \ln t$ into the equation of a straight line ($y = mx + c$).

$$\ln t = -1.2 \ln h + c$$

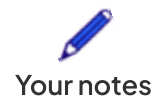
Substitute either coordinate in and rearrange to find c .

$$\begin{aligned} -0.9 &= -1.2(1) + c \\ c &= -0.9 + 1.2 \\ c &= 0.3 \end{aligned}$$

Compare this to the linearised form of $t = ah^b$. Take logarithms of both sides first and rearrange if you can't remember the correct form.

$$\begin{aligned} \ln t &= \ln(ah^b) \\ \ln t &= \ln a + b \ln h \end{aligned}$$

So b is the value in front of $\ln h$ and $\ln a = 0.3$. Solve $\ln a = 0.3$ by taking e of both sides.



$$\ln a = 0.3$$

$$e^{\ln a} = e^{0.3}$$

$$a = e^{0.3}$$

$$a = e^{0.3}, b = -1.2$$

Transforming Relationships in the Form $y = Ab^x$

How do I use logarithms to linearise a graph in the form $y = A(b^x)$?

- Logarithms can be used to linearise graphs of **exponential functions**
- Suppose $y = Ab^x$
 - You can take logarithms of both sides
 - $\log y = \log (Ab^x)$
 - You can split the right hand side into the sum of two logarithms
 - $\log y = \log A + \log (b^x)$
 - You can bring down the power in the final term
 - $\log y = \log A + x \log b$
- $\log y = \log A + x \log b$ is in linear form $Y = mX + c$
 - $Y = \log y$
 - $X = x$
 - $m = \log b$
 - $c = \log A$

How can I use linearised form to find the unknown constants?

- After linearising the function it will be in the form $\log y = \log A + x \log b$
 - $\log b$ is the gradient of the straight line graph (m)
 - $\log A$ is the y -intercept of the straight line graph (c)



Your notes

- You will need to find the value of A by solving the equation $\log A = c$
 - The value of c will either be given or will need to be found
- You will need to find the value of b by solving the equation $\log b = m$
 - The value of m will either be given or will need to be found



Examiner Tips and Tricks

- Unless the question specifies, you can choose whether to use $\ln$, $\lg$ or $\log$
- Remember you will need to solve the equation at the end
 - If using $\lg$, solve by taking 10 to the power of each side
 - If using $\ln$, solve by taking e to the power of each side



Worked Example

Variables X and y are such that when $\lg y$ is plotted against X , a straight line passing through the points $(2, 5)$ and $(5, 8)$ is obtained.

Show that $y = A \times b^x$ where A and b are constants to be found.

Find the gradient of the straight line between the two coordinates.

$$m = \frac{8 - 5}{5 - 2} = \frac{3}{3} = 1$$

Substitute $m = 1$, $X = x$ and $Y = \lg y$ into the equation of a straight line ($Y = mX + c$).

$$\lg y = x + c$$

Substitute either coordinate in and rearrange to find c .

$$5 = 2 + c$$

$$c = 3$$

$$\lg y = x + 3$$



Your notes

Linearise $y = A \times b^x$.

Take logarithms of both sides first and rearrange if you can't remember the correct form. Take logarithms of both sides.

$$\lg y = \lg (Ab^x)$$

Split the right-hand side into the sum of two logarithms.

$$\lg y = \lg A + \lg b^x$$

Bring down the power in the final term.

$$\lg y = \lg A + x \lg b$$

Compare this to the equation of the line.

$$\lg b = 1 \text{ and } \lg A = 3$$

Solve $\lg b = 1$ by raising 10 to the power of both sides.

$$\lg b = 1$$

$$10^{\lg b} = 10^1$$

$$b = 10$$

Solve $\lg A = 3$ by raising 10 to the power of both sides.

$$\lg A = 3$$

$$10^{\lg A} = 10^3$$

$$A = 1000$$

$$y = 1000 \times 10^x$$